Business Analytics with R – Regression Analysis

Outline

- Goals of this lecture
- What is a (linear) regression analysis?
- How to perform a regression analyis in R?

Goals of the current lecture

- Students know the basic principles of OLS regression analysis
- Students can apply regression analyzes in R.
  - Estimating and interpreting Linear Regression (Ordinary Least Squares)
  - Can use the estimated regression model to make predictions
  - Can evaluate the quality of model predictions
  - Know how to perform a regression diagnostic

What is Linear Regression?

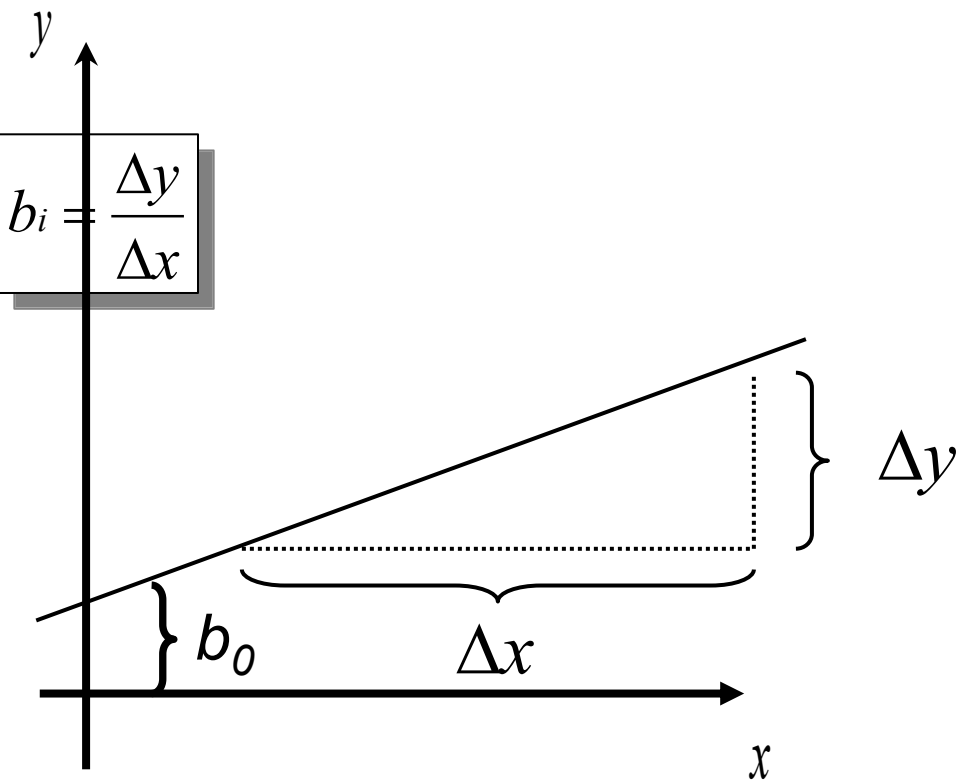
A way of predicting the value of one variable from another.

It is a hypothetical model of the relationship between two variables.

The model used is a linear one.

Therefore, we describe the relationship using the equation of a straight line.

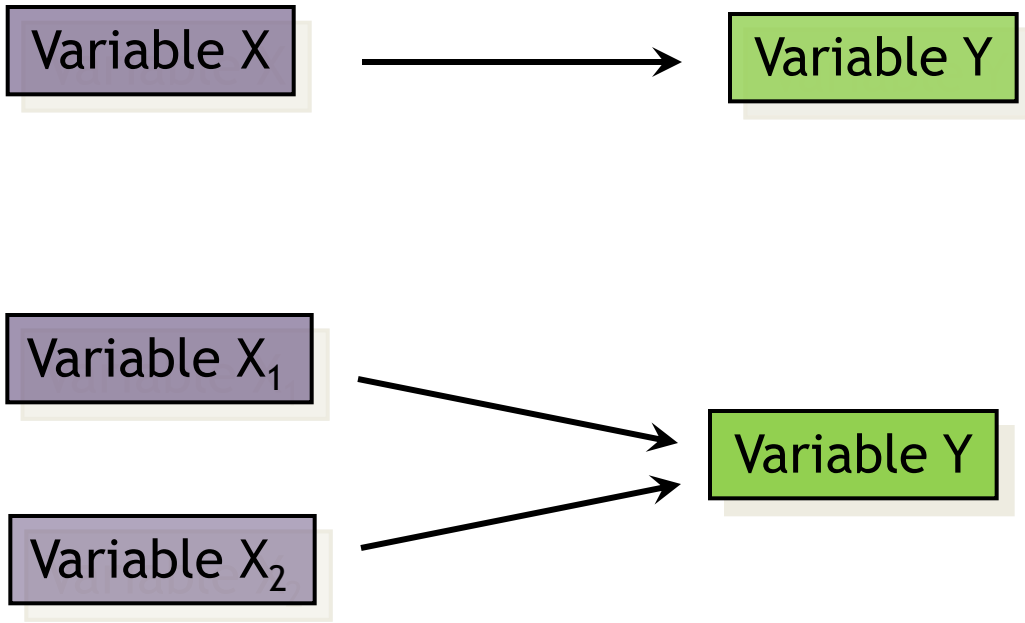
Describing a line



- b_i
  - Regression coefficient for the predictor
  - Gradient (slope) of the regression line
  - Direction/Strength of Relationship
- b_0
  - Intercept (value of Y when X = 0)
  - Point at which the regression line crosses the Y-axis (ordinate)

$$Y_i = b_0 + b_1 X_i + \varepsilon_i$$

Linear Regression (OLS)



SIMPLE LINEAR REGRESSION

One independent Variable:

$Y=f(X)$

MULTIPLE LINEAR REGRESSION

$Y=f(X_1, X_2, \ldots, X_j, \ldots, X_J)$

Regression Model: We make errors

$$Y_i = b_0 + b_1 X_i + \varepsilon_i$$

Error / Residual

$$\hat{Y}_i = b_0 + b_1 X_i$$

Predicted value of dependent variable

The Method of Least Squares

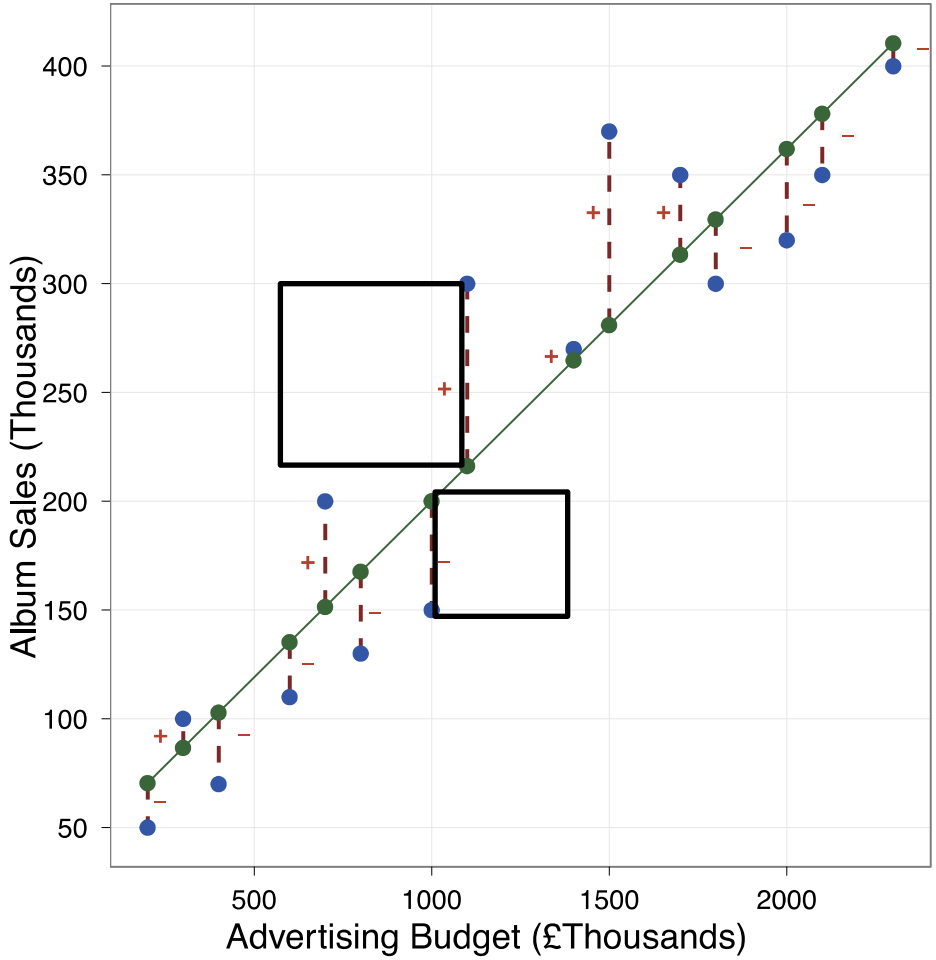


FIGURE 8.4
A scatterplot of some data with a line representing the general trend. The vertical lines (dotted) represent the differences (or residuals) between the line and the actual data

How well does the Model fit the data?

There are two ways to assess the accuracy of the model in the sample:

Residual Statistics

Standardized Residuals

Influential cases

Cook's distance

R and R^2

R

The correlation between the observed values of the outcome, and the values predicted by the model.

R^2

The proportion of variance accounted for by the model.

Adj. R^2

Controlled for the number of predictors – *R2 increases with an increasing number of predictors*

Estimating a regression model in R

```
###Now we estimate a multivariate model with two predictors.
model <- lm(totalRent ~ livingSpace + yearConstructed + balcony, data = df)

##Displaying the model with stargazer
stargazer(model, type = "text")
```

Estimating a regression model in R

```
==============================================
                    Dependent variable:
                ---------------------------
                       totalRent
----------------------------------------------
livingSpace              5.729***
                         (0.082)

yearConstructed           0.015
                         (0.028)

balconyy                13.160***
                         (2.904)

Constant                 51.512
                        (55.697)

----------------------------------------------
Observations              1,096
R2                       0.822
Adjusted R2               0.822
Residual Std. Error    43.701 (df = 1092)
F Statistic        1,681.918*** (df = 3; 1092)
==============================================
Note:              *p<0.1; **p<0.05; ***p<0.01
```

Predictions on test data

###Predicting for test data and storing predictions as new variable in test data frame

```
test$prediction <- predict(model3, newdata= test)

test$prediction
```

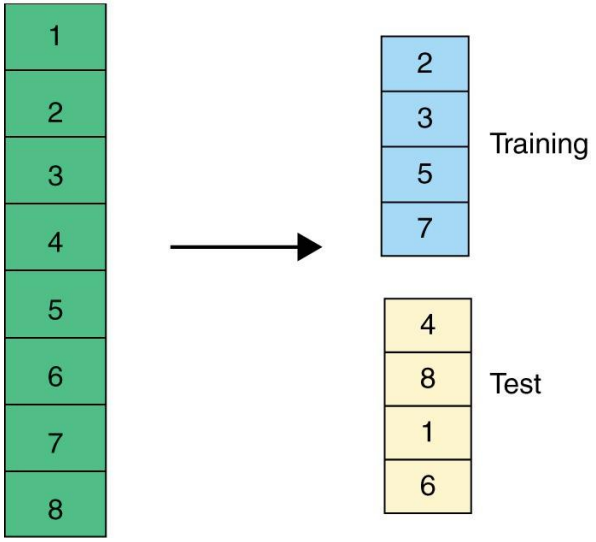
Measures of model Accuracy

```
##Calculating measures of model accuracy

###calculating mape
test <- test %>%
  mutate(MAPE = abs(prediction - totalRent)/totalRent)

mean(test$MAPE)

###Calculating Sales Ratios
test <- test %>%
  mutate(SR = prediction / totalRent)

mean(test$SR)
```

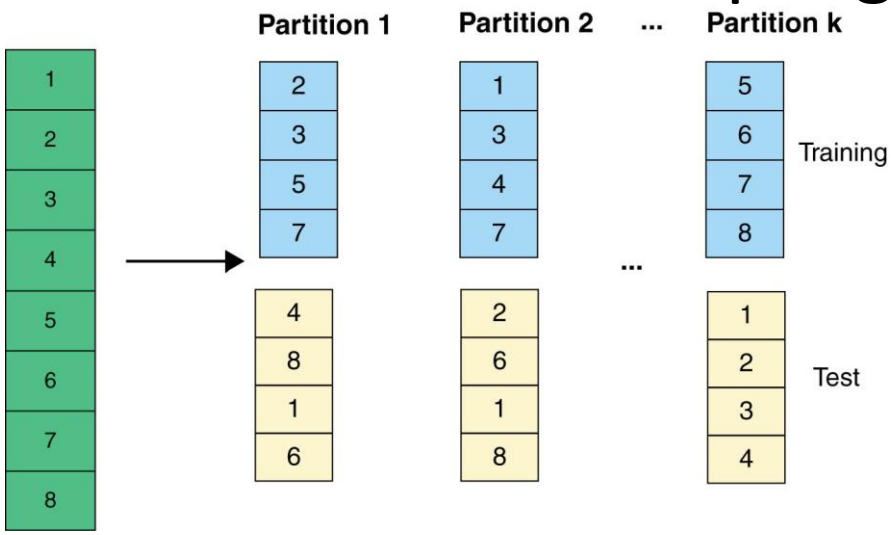
Model Validation and Performance Generalisation

Model validation & Performance Generalisation

Holdout validation

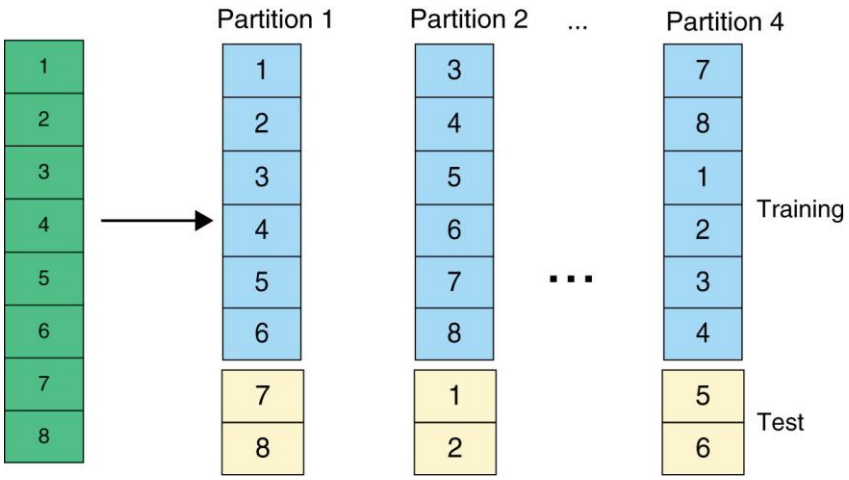


Random sub-sampling

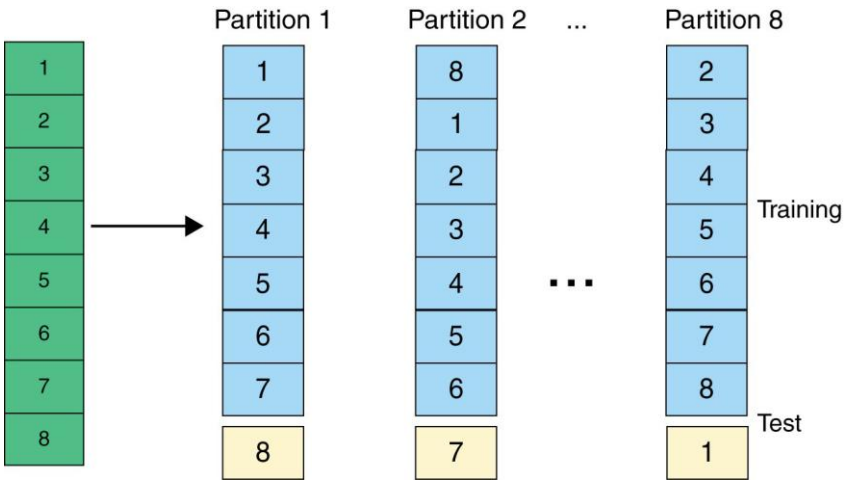


Model validation & Performance Generalisation

k-fold cross validation



Leave-one-out



Model validation & Performance Generalisation

```
library(caret)

# Define number of folds
k <- 3

# Create folds
folds <- createFolds(df.numeric.variables$totalRent, k = 3, list = FALSE)

for(i in 1:k){
  # We keep fold i for testing the model and use all other folds for training the model.
  test <- df.numeric.variables%>%
    filter(folds==i)
  train <- df.numeric.variables%>%
    filter(folds!=i)
  # Training model
  model <- lm(totalRent~livingSpace+yearConstructed,data=train)

cat("Model summary for fold",i,":\n")
  print(summary(model))
}
```

Questions?